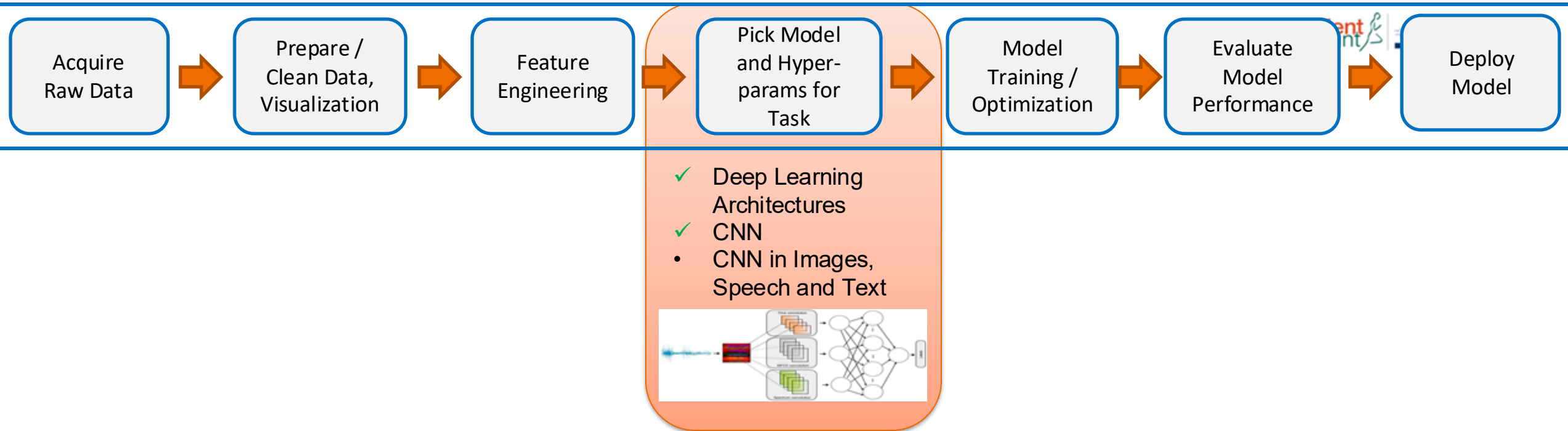
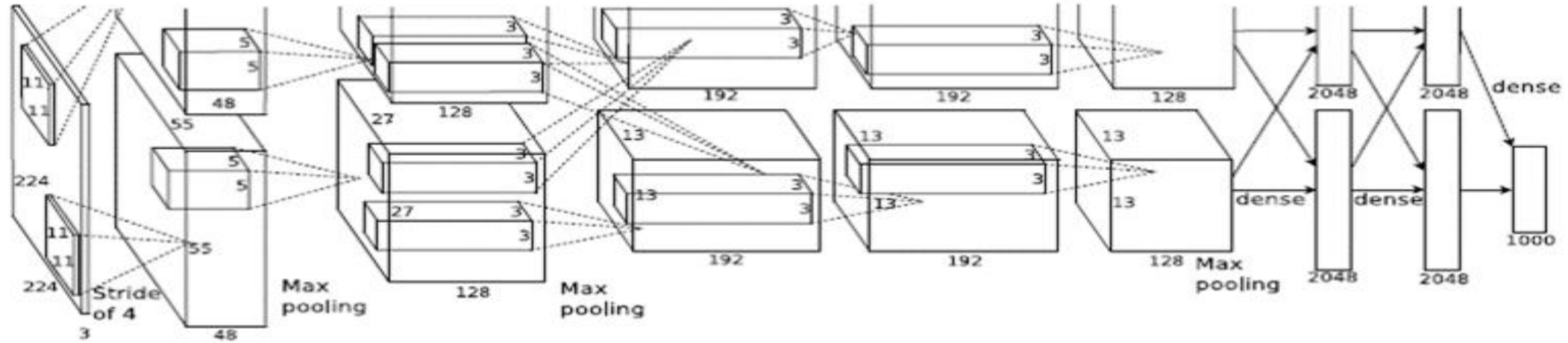




CNN in Images, Speech and Text

“CNN for ALL”

Recap: Turning Point: AlexNet



ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
University of Toronto
ilya@cs.utoronto.ca

Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

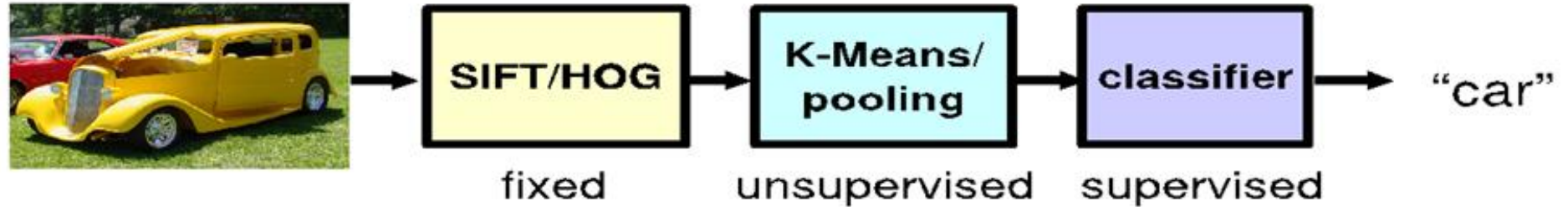
ImageNet Classification Task:

Previous Best : ~25% (CVPR-2011)

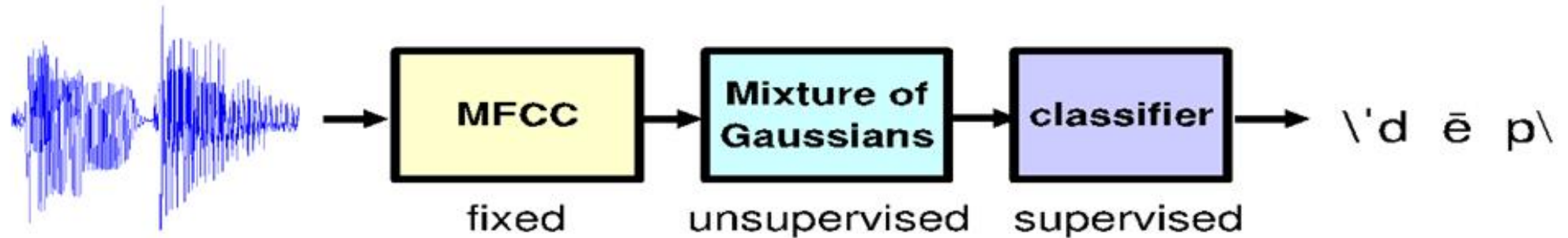
AlexNet : ~15 % (NIPS-2012)

Common Pipeline: Till Then

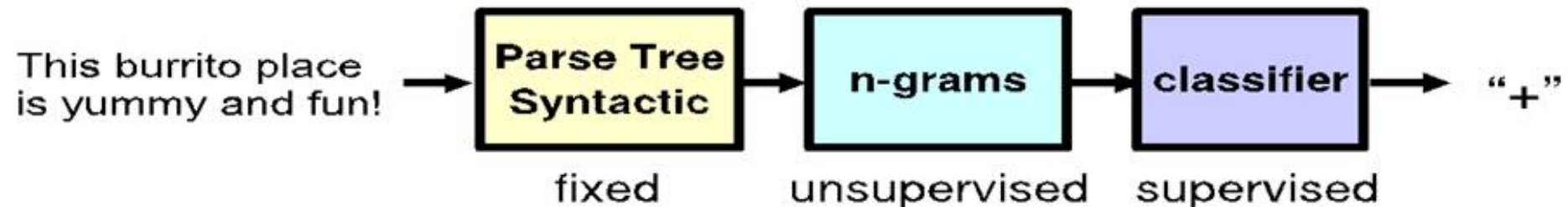
- VISION:



- SPEECH:



- NLP:



Learn the full pipeline

- **VISION:**

- Pixels → edge → texon → motif → part → object

- **SPEECH:**

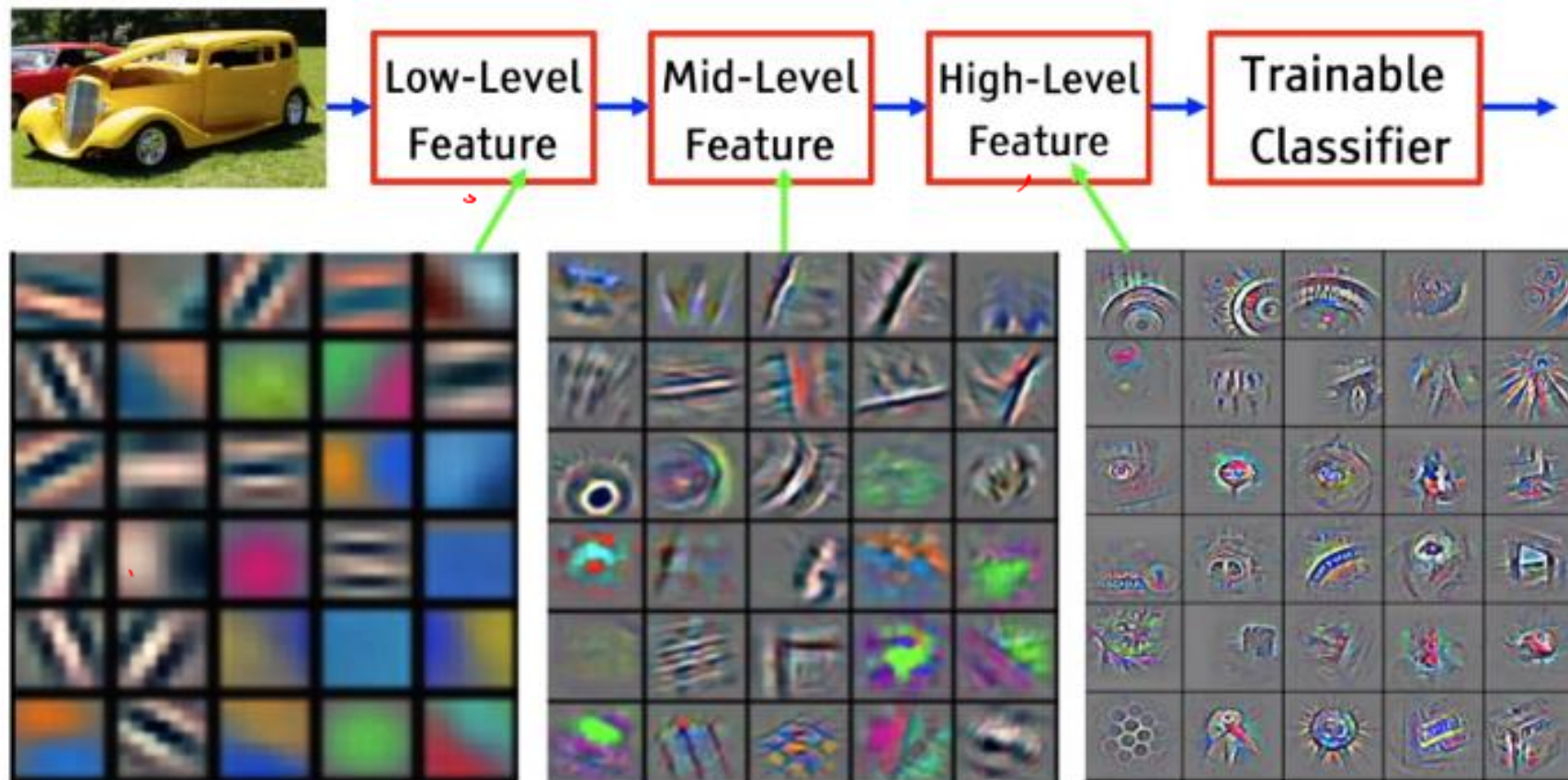
- Sample → spectral → band → formant → motif → phone → word

- **NLP:**

- Character → word → NP/VP/.. → Clause → sentence → story

Deep Learnt Features

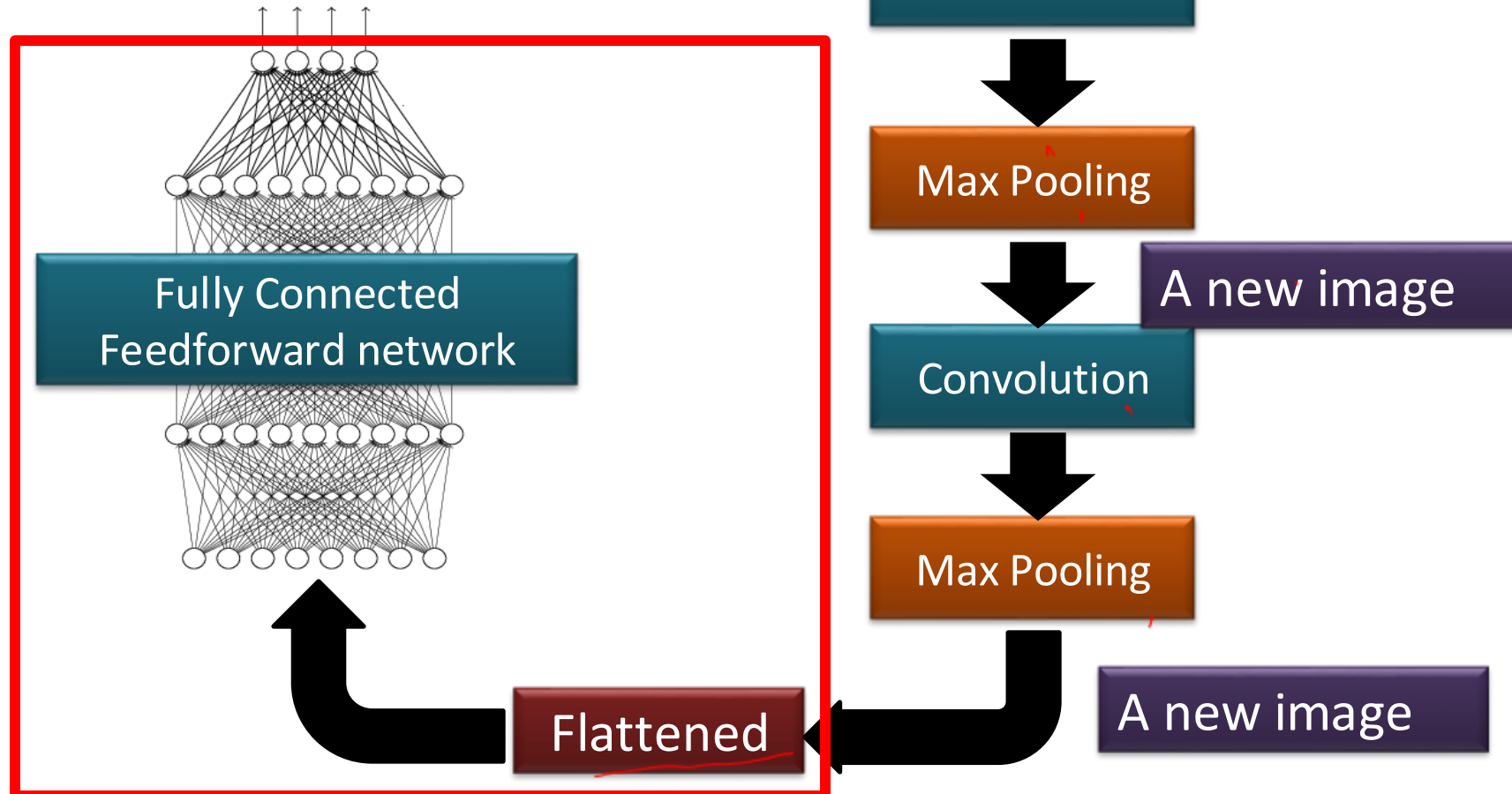
- It's **deep** if it has **more than one stage** of non-linear feature transformation.



The whole CNN

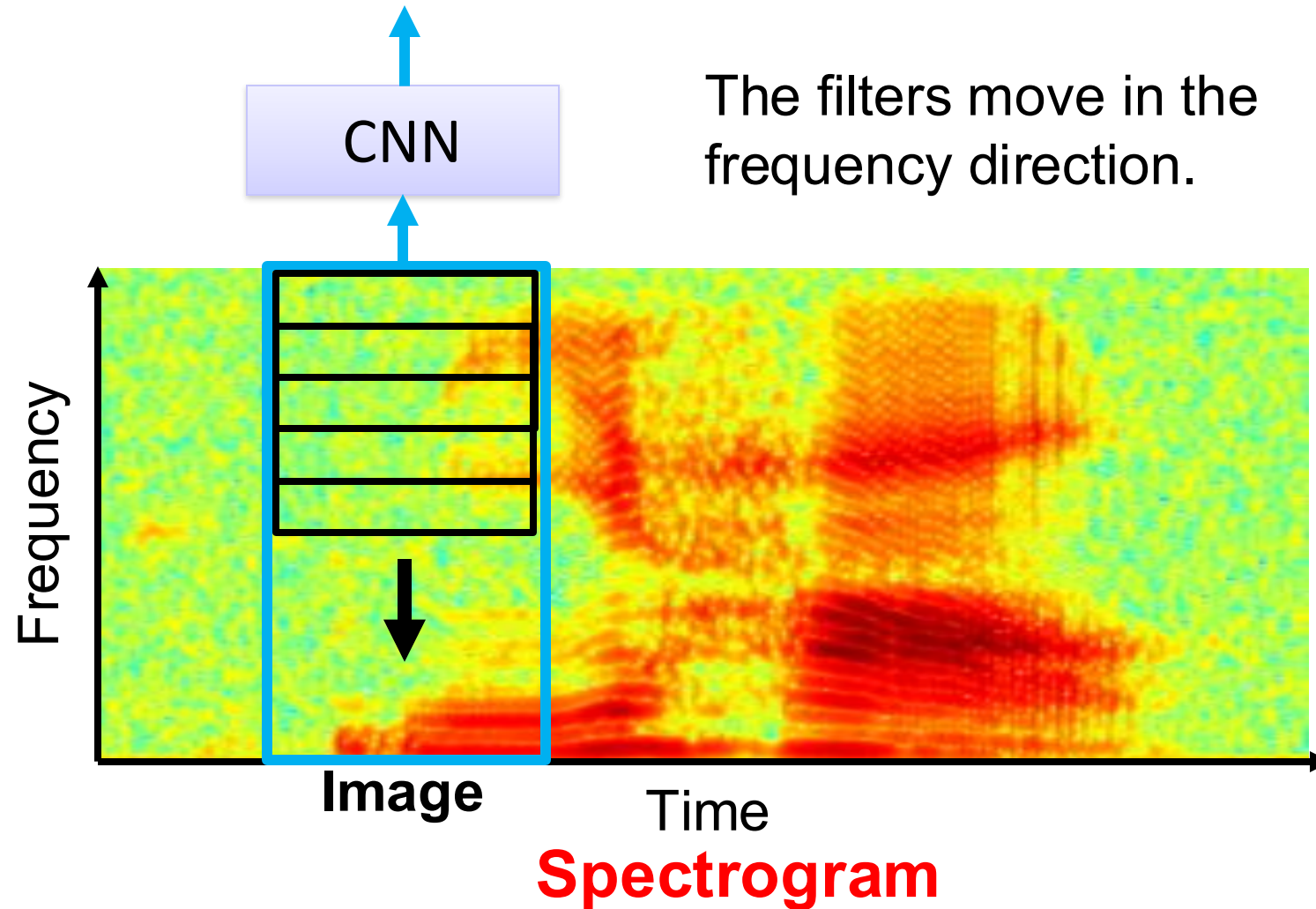


cat dog

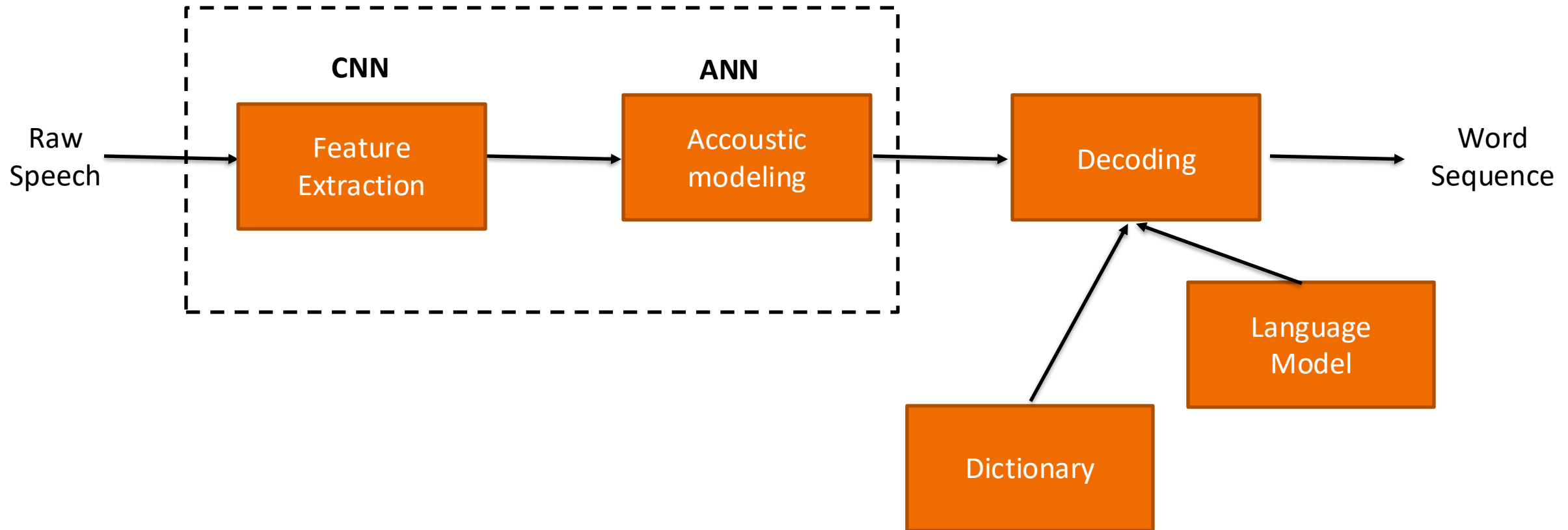


BLANK SLIDE

CNN in speech recognition

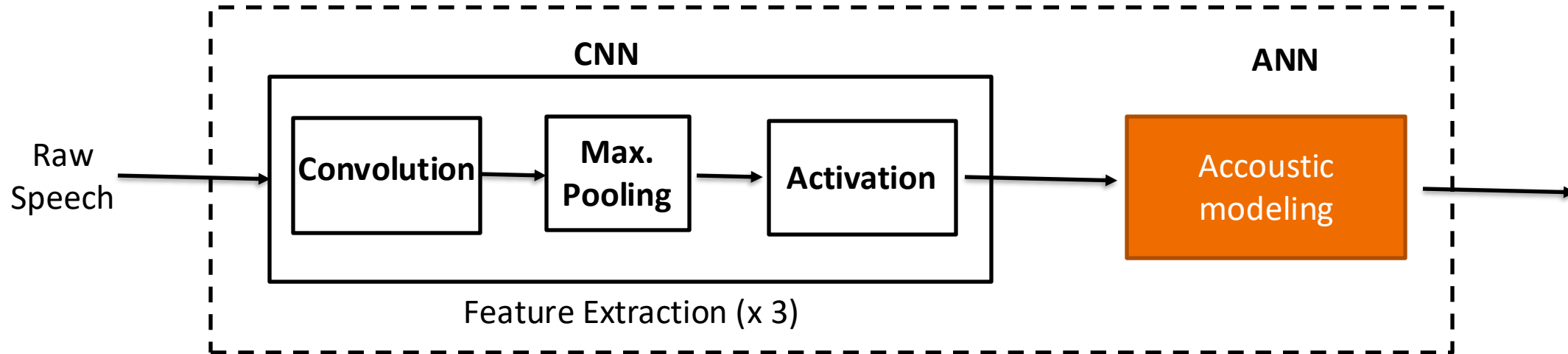


Speech Recognition model

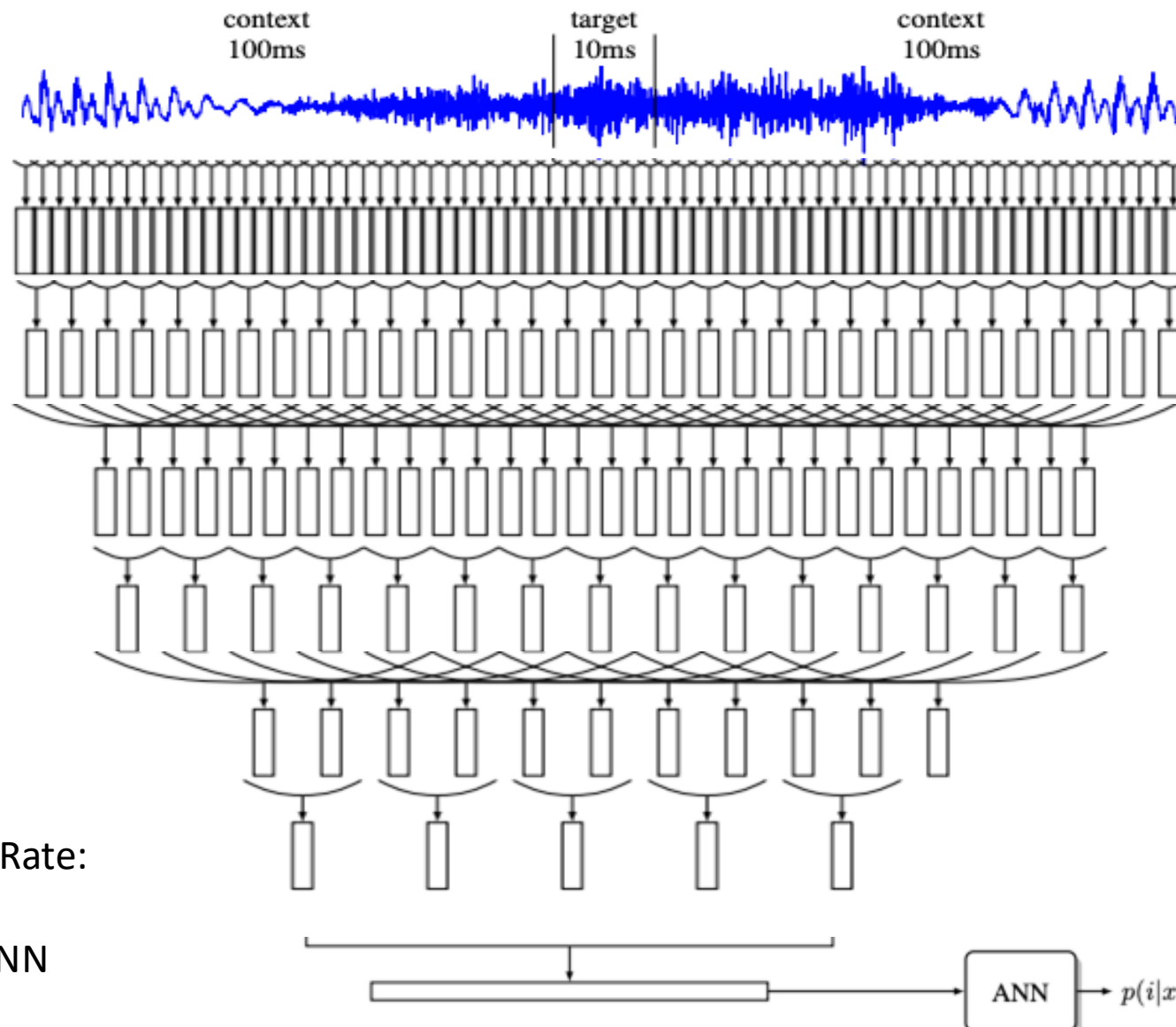


Dimitri Palaz, Mathew Magimai-Doss and Ronan Collobert, "Convolutional Neural Networks-based Continuous Speech Recognition using Raw Speech Signal", ICASSP 2015, pp.4295-4299.

Speech Recognition model



Detailed View



Convolution

Max pooling

Convolution

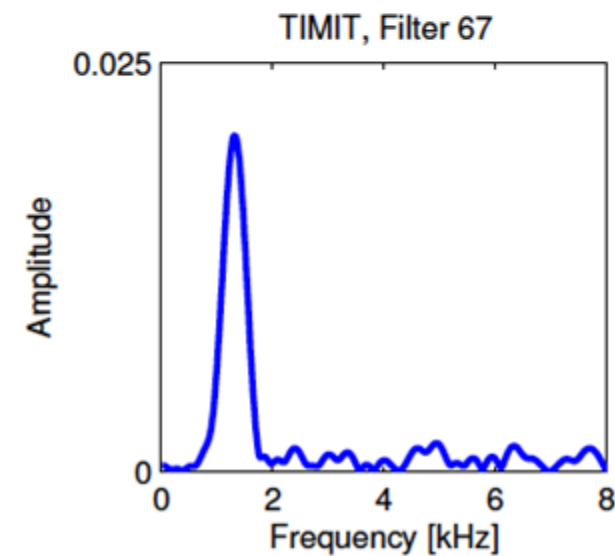
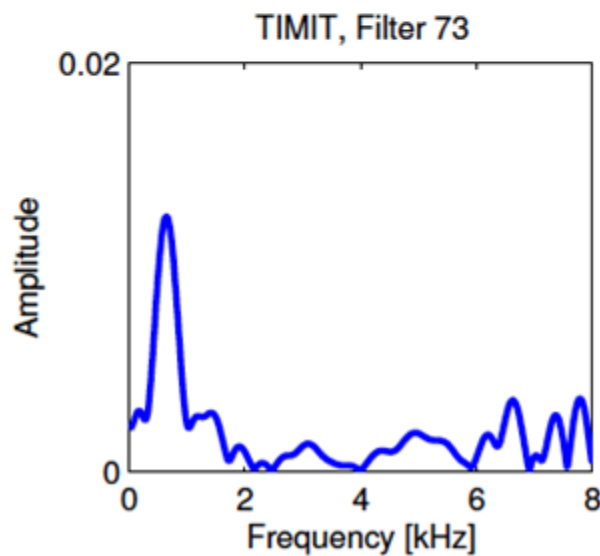
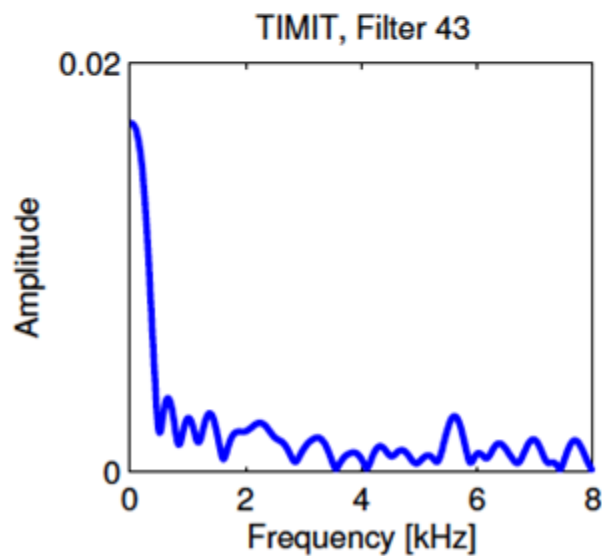
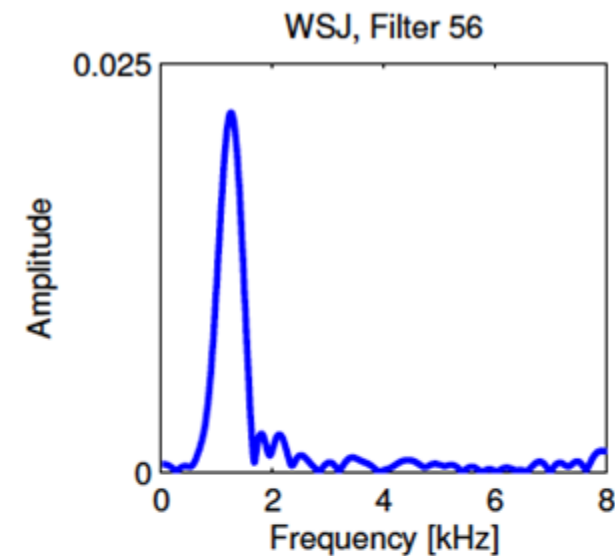
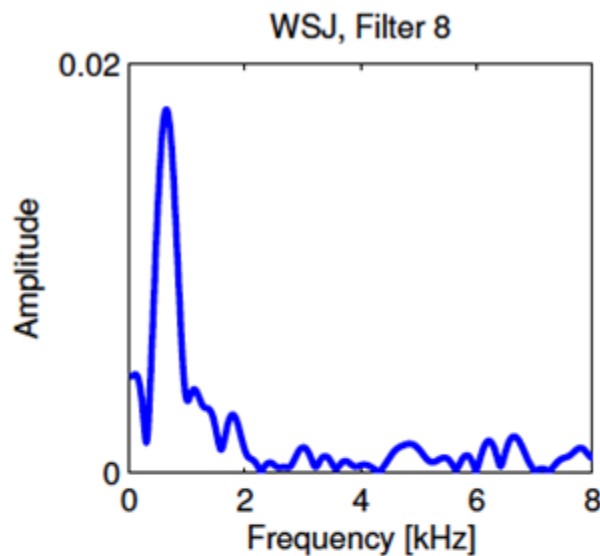
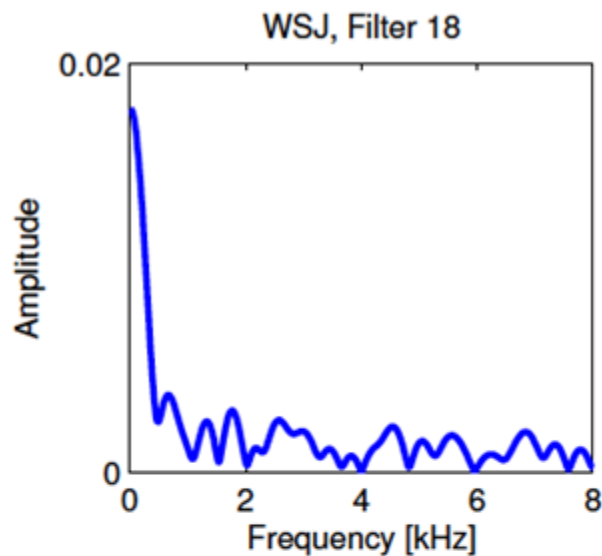
Max pooling

Convolution

Max pooling

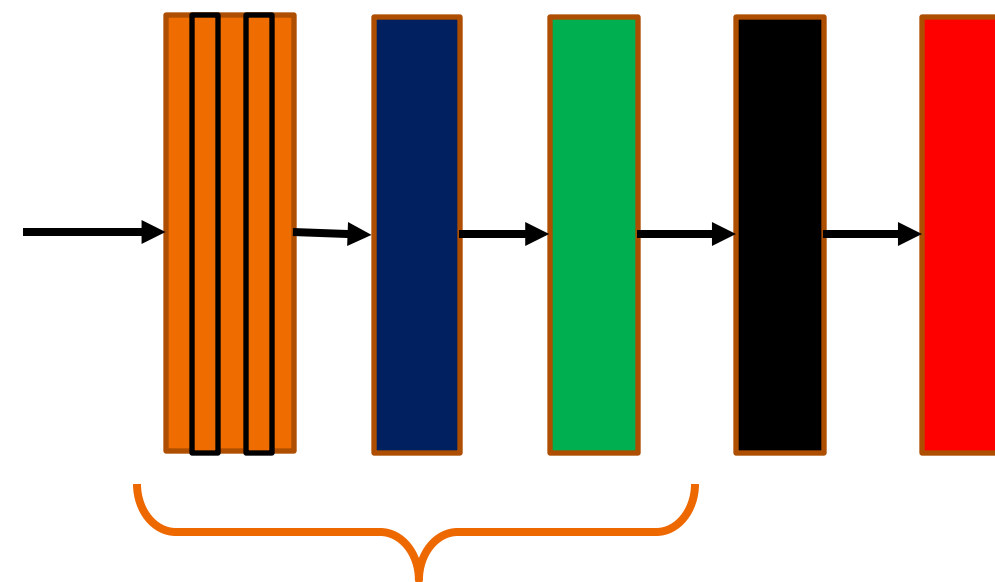
WSJ Word Error Rate:
64.% vs 5.6% for
MFCC+ANN vs CNN

Comparing Learnt Filters: WSJ vs. TIMIT



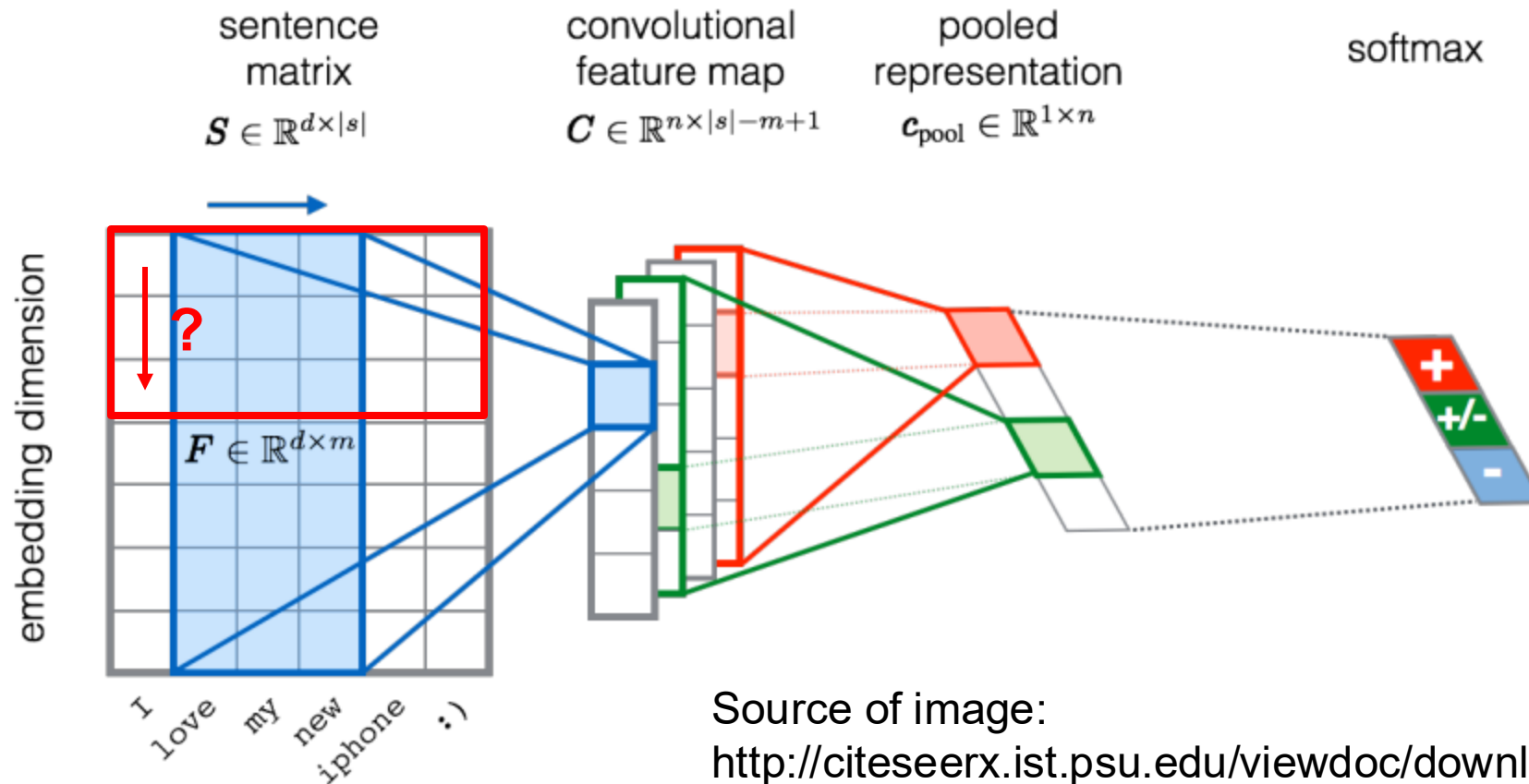
Summary: Layers of a Neural Network

- Based on the connection pattern and operations, we can think of a layer in a Neural Network as:
 - Convolution
 - A Layer can have multiple Channels
 - Non-Linear (often not drawn)
 - Max-Pooling
 - Fully Connected
 - Soft Max



This is often repeated
multiple times

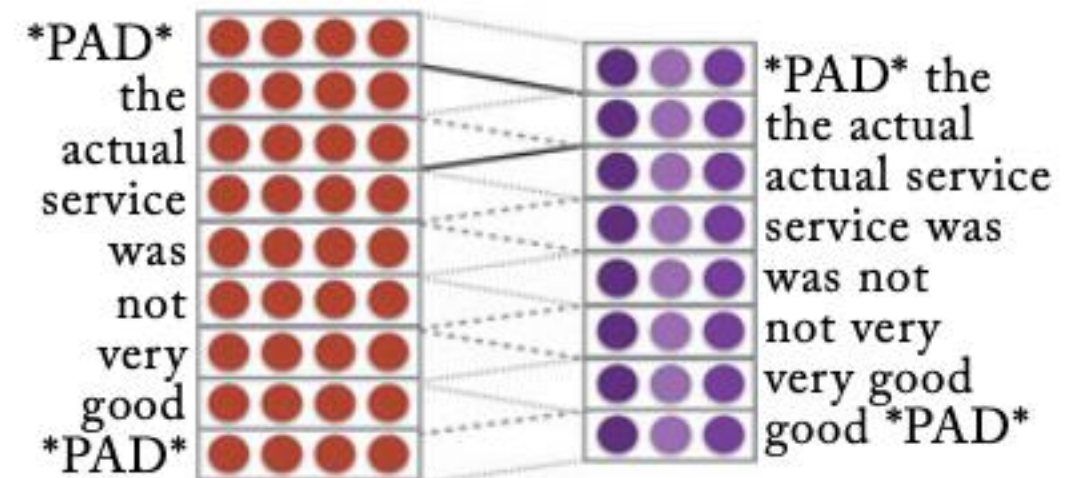
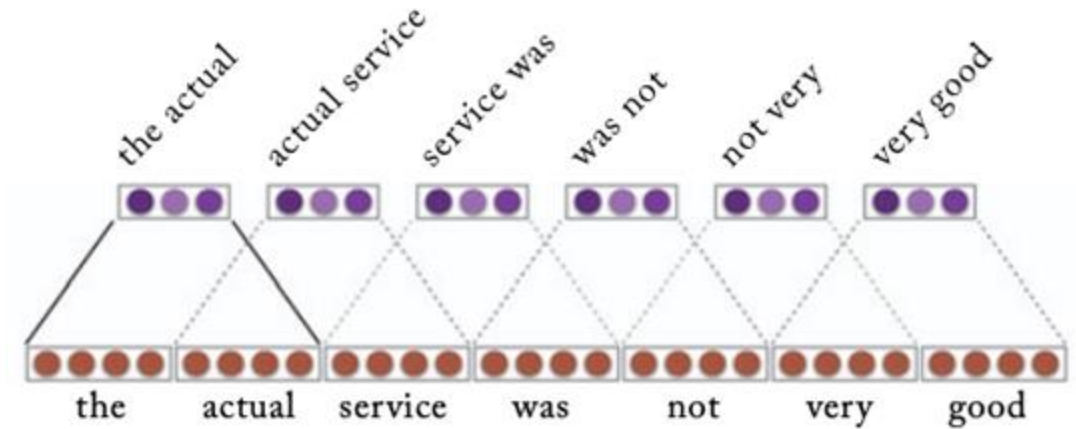
CNN in text classification



Source of image:
<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.703.6858&rep=rep1&type=pdf>

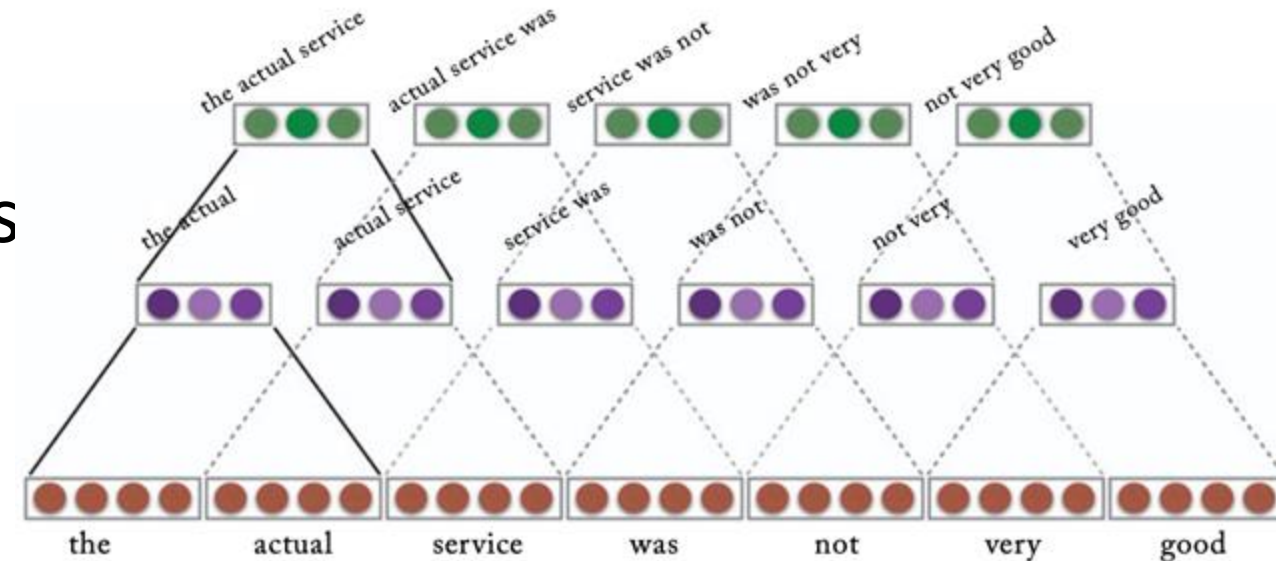
Convolution

- n-word sliding window over the sentence
- Learning to identify indicative **ngrams** in the input
- Filter transforms a window of k words into a scalar value
- Several filters applied to capture properties of the words in the window



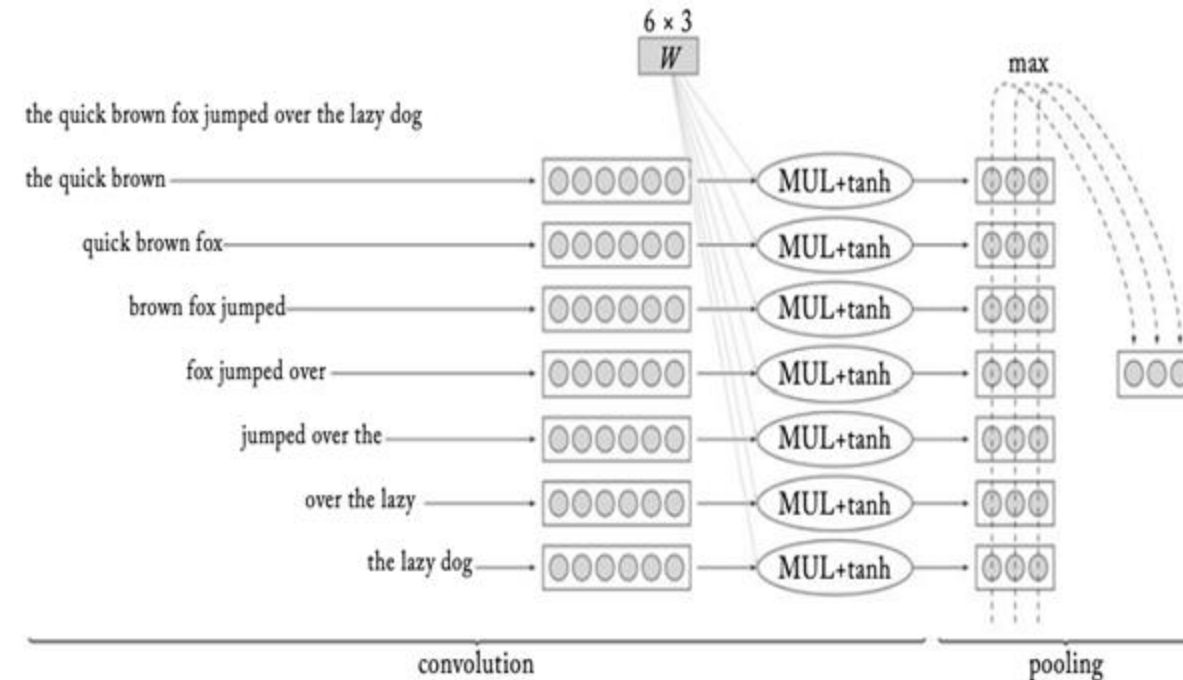
HIERARCHICAL CONVOLUTIONS

- This approach can be extended to hierarchy of convolutional layers
- Increase in depth of CNN leads to capture increasingly larger effective windows for a sentence
- Dilated convolutions help in learning the relationship between Non-Adjacent words

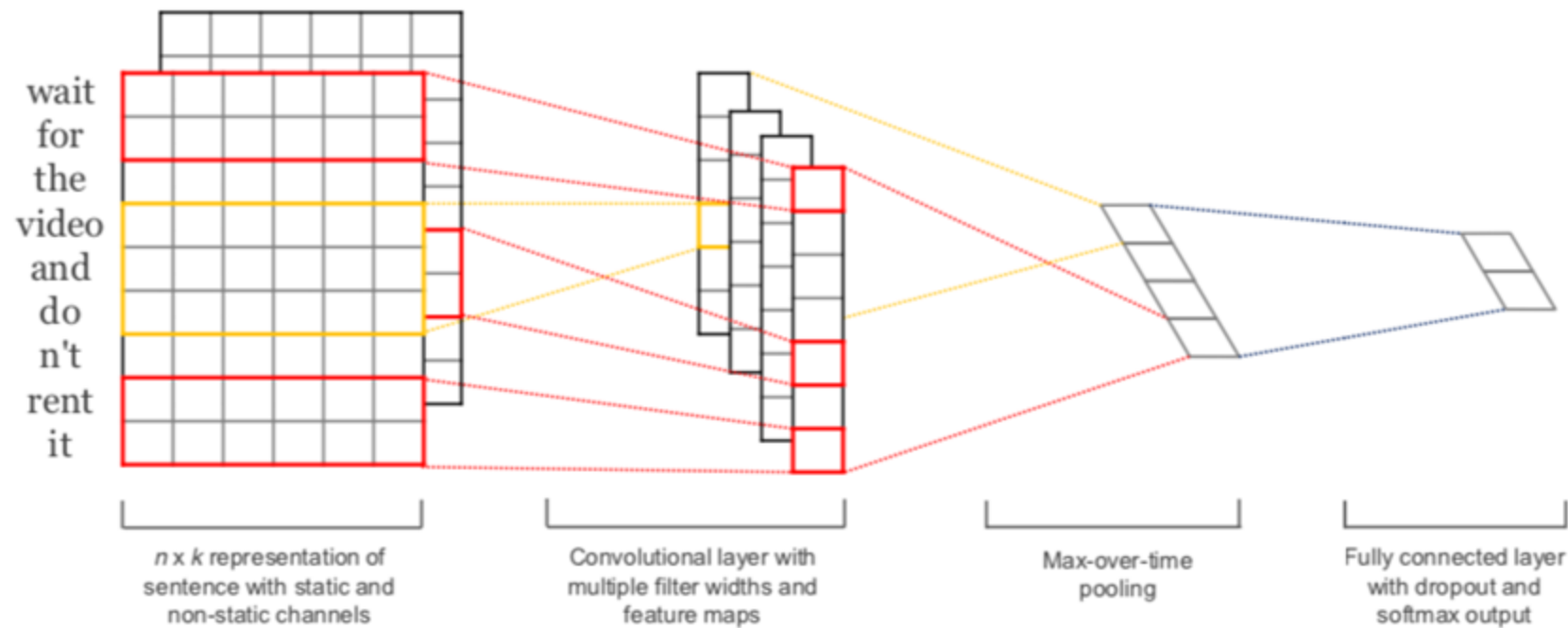


Pooling

- Pooling operation combines the vectors from the different windows into a single dimensional vector
- By taking the max or the average value
- The intention is to focus on the most important features
- Each filter extracts a different indicator from the window
- Pooling operation zooms in on the important indicators



Convolutional Neural Networks for Sentence Classification



Convolutional Neural Networks for Sentence Classification, Yoon Kim

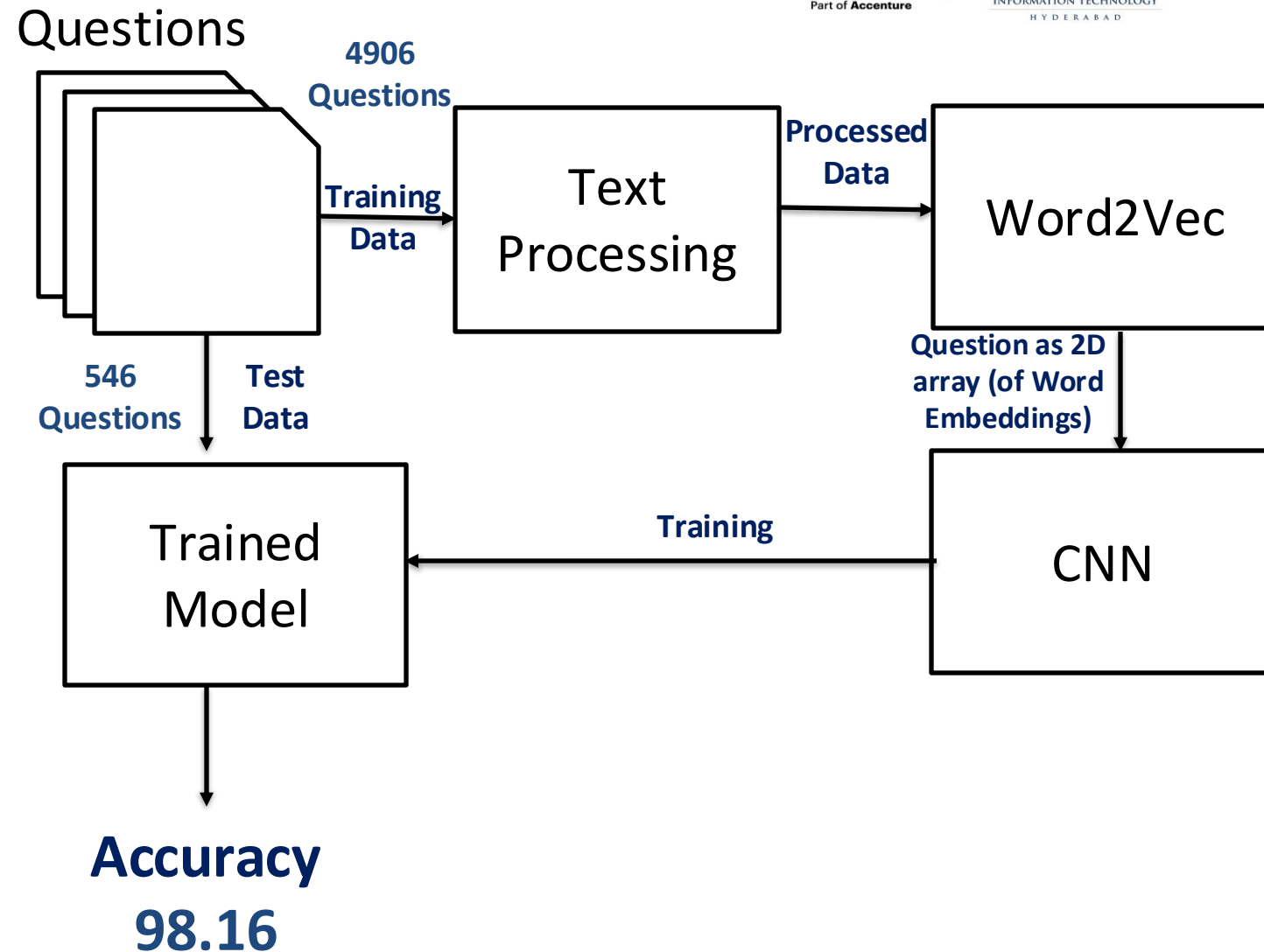
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Case Study

Text Classification with CNNs

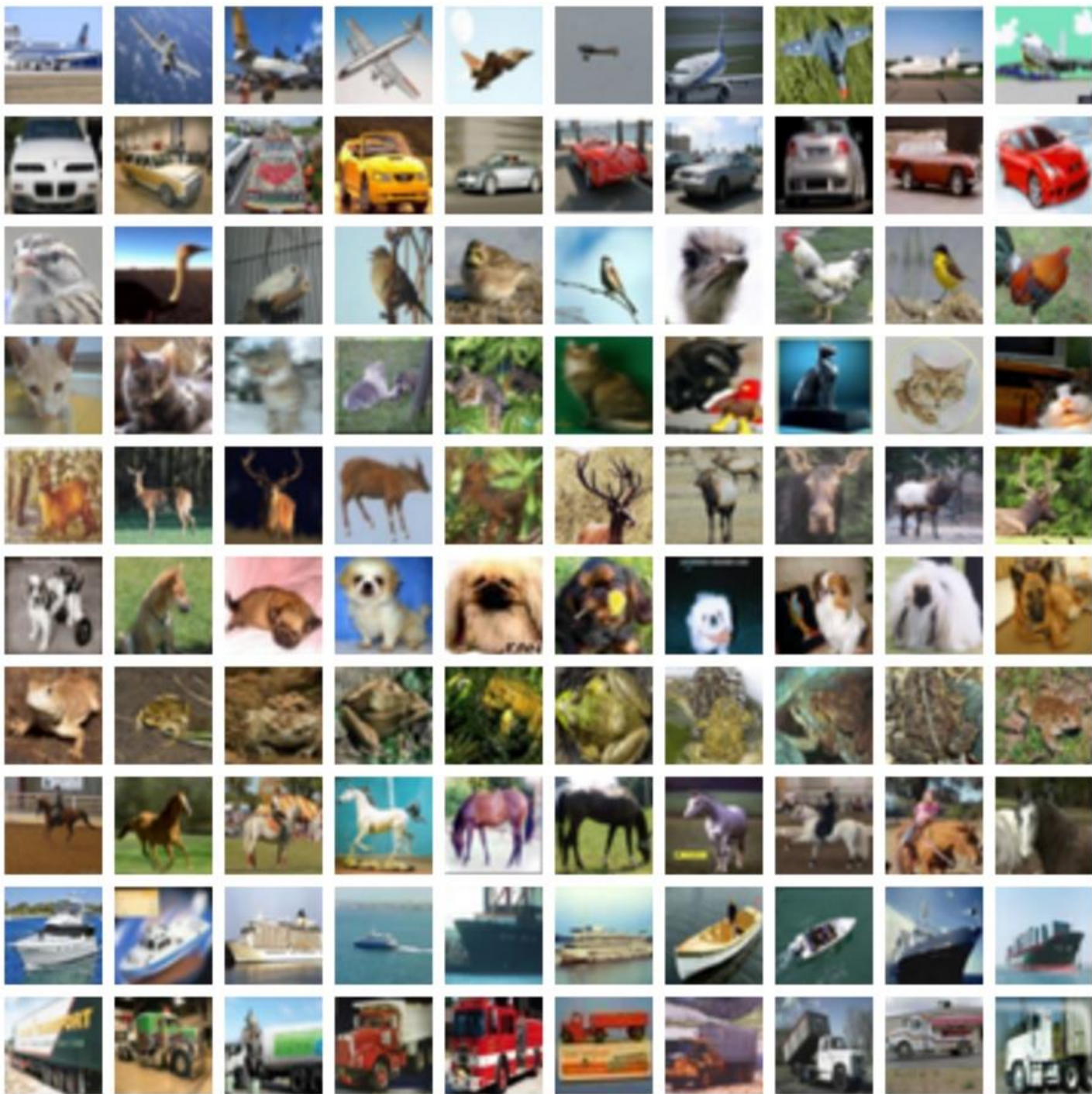
NLP with CNN

- Problem: Classification of Questions
 - Data: Various questions labeled based on the type
 - Labels: Abbreviation, Entity, Description, Human, Location and Numeric value.
- Process the data: Remove punctuations etc.
- Represent the Sentence as a 2D array with word embeddings
- Train the CNN
- Use the trained model for Testing



Is there an inherent hierarchy in natural images?

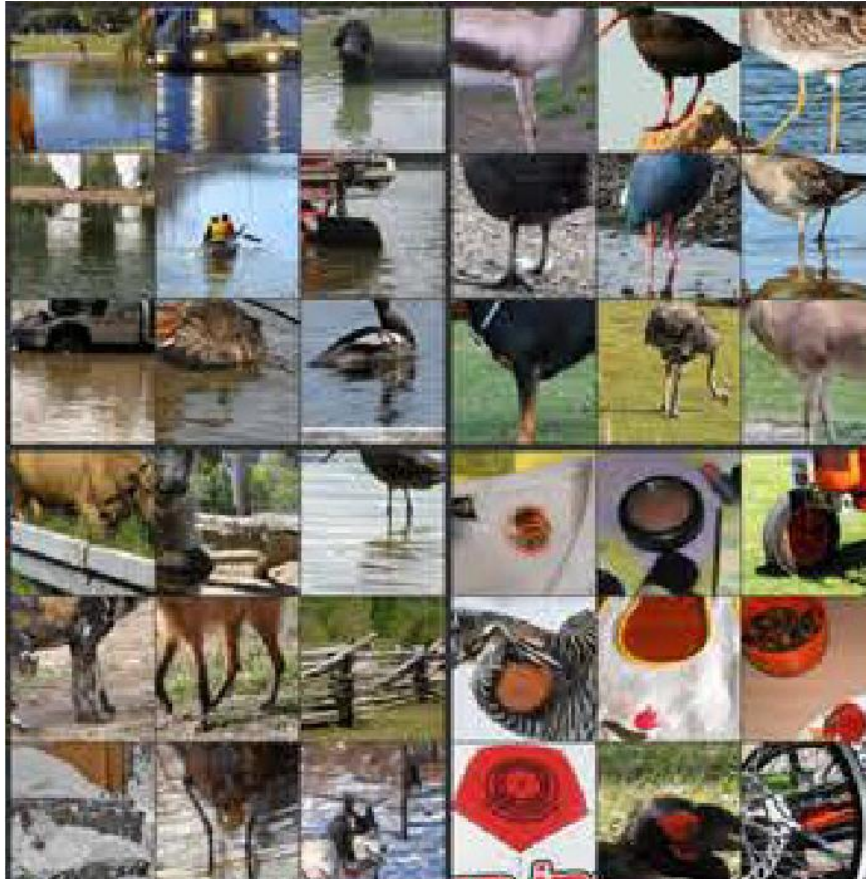




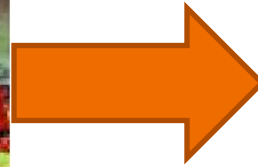
Can you identify any structure or common elements in these images?

CIFAR10 dataset
Image source:
[CIFAR-10 image classification with Keras ConvNet - Giuseppe Bonaccorso](#)

Hierarchy of visual elements



Parts of images: one step down the hierarchy



Even lower down the hierarchy

Image source:
Zeiler, Matthew D., and Rob Fergus. 2014. "Visualizing and Understanding Convolutional Networks." *ECCV 2014*,

Even lower down the
hierarchy:

Textures and colours

Image source:
Zeiler, Matthew
D., and Rob
Fergus. 2014.
“Visualizing
and
Understanding
Convolutional
Networks.
ECCV 2014,

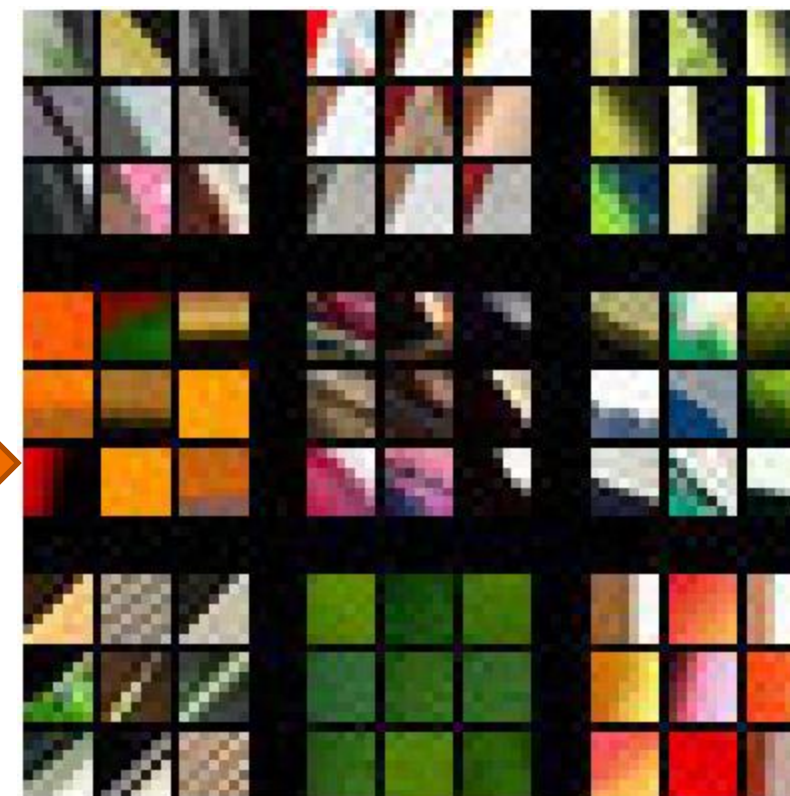
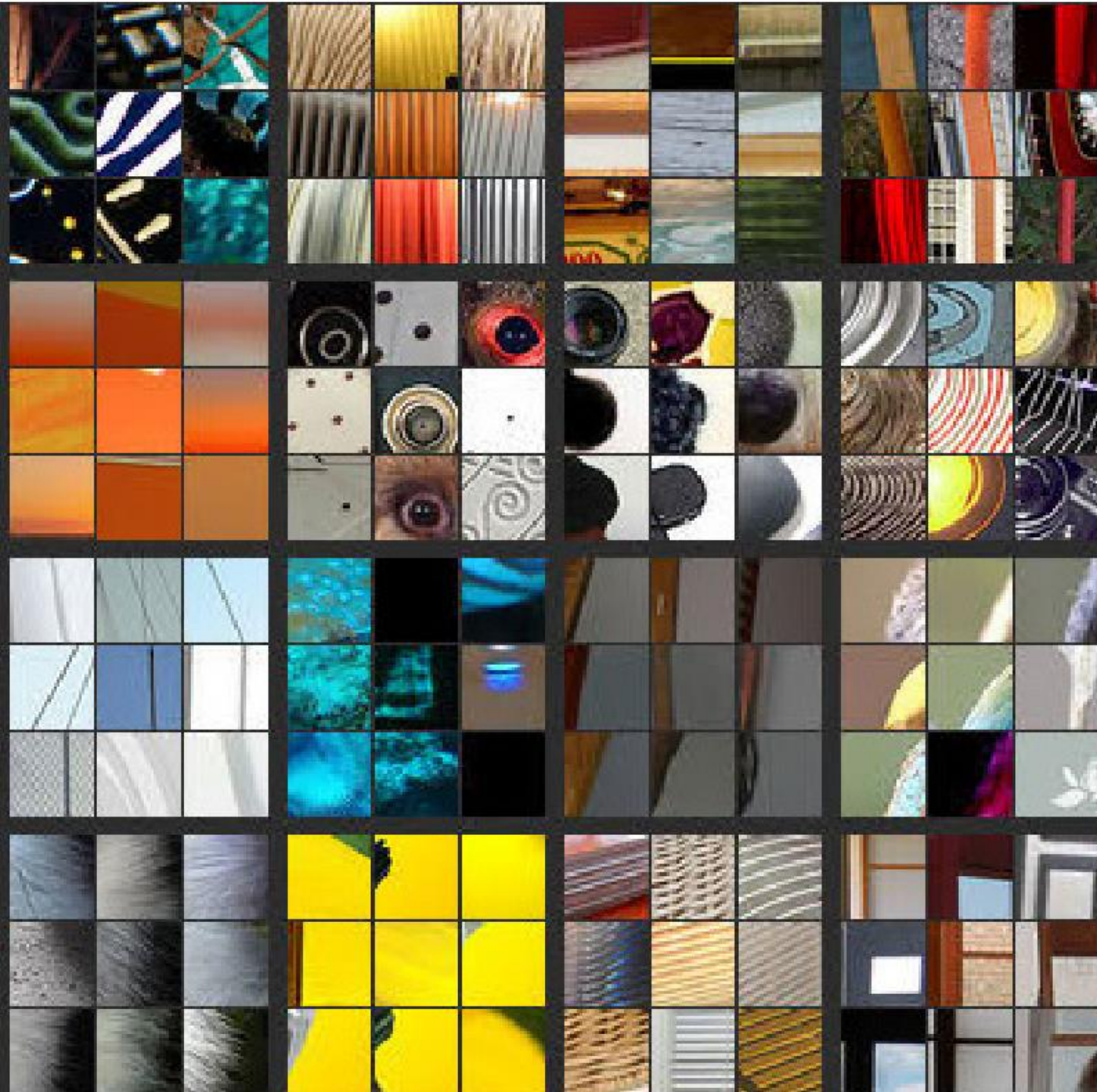


Image source:
Zeiler, Matthew D., and Rob
Fergus. 2014. "Visualizing and
Understanding Convolutional
Neural Networks." ECCV 2014.

Hierarchy in face images?

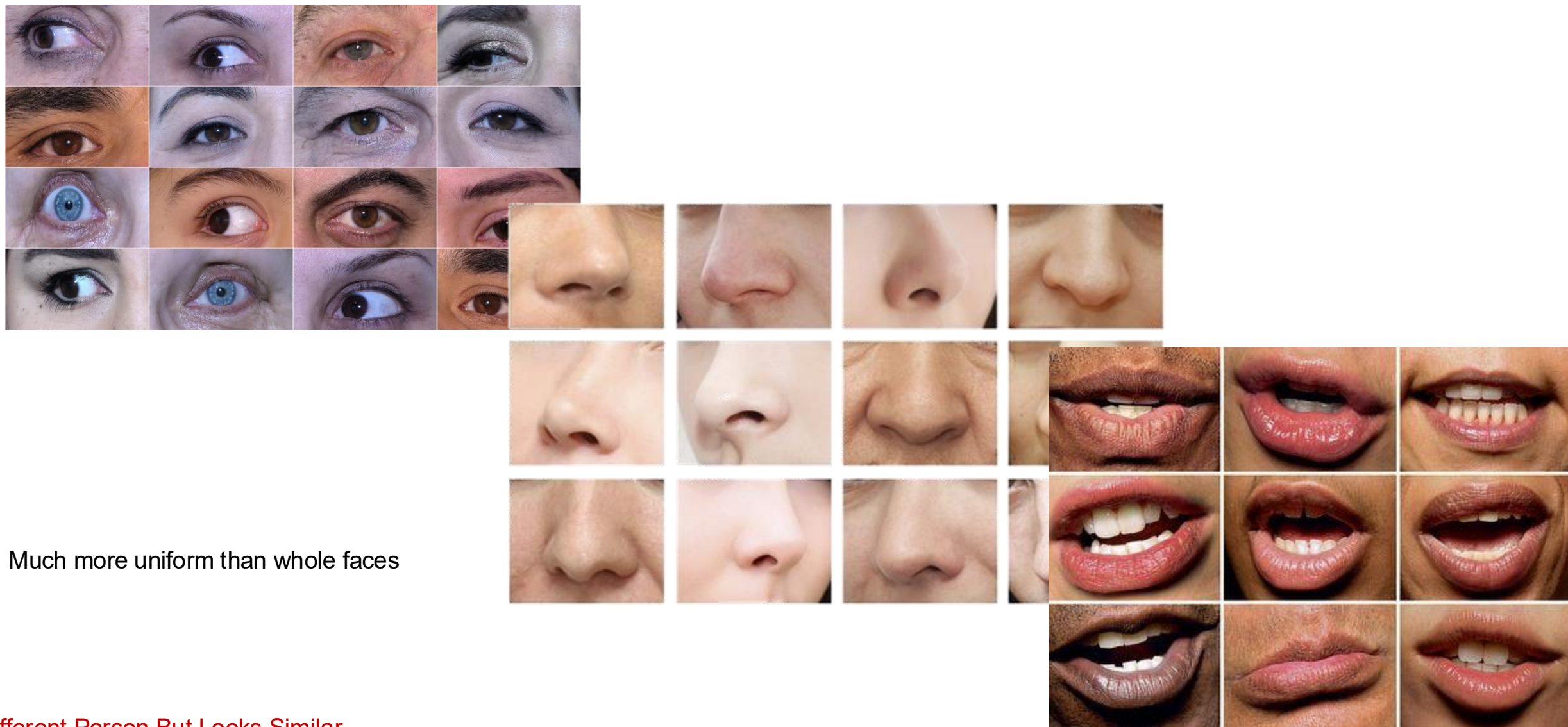


Highest level: whole faces, identities



Same Person still Looks Different

Face parts



Much more uniform than whole faces

Different Person But Looks Similar

Still Smaller: Textures? Even more uniform



Do CNNs work in a similar way?

Convolution as Part Search

Higher level filters

- If we create a filter for each pattern we want to recognize, there will be too many variations!
- Utilize image hierarchy

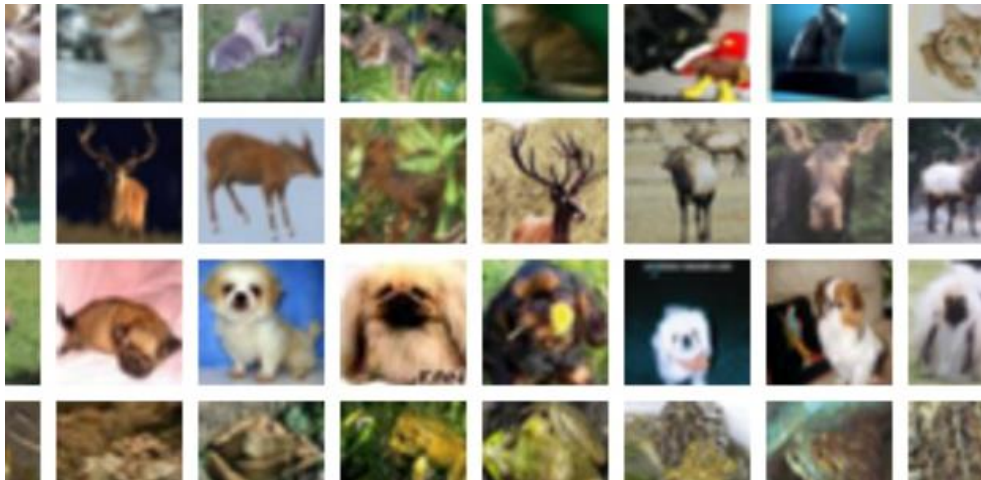
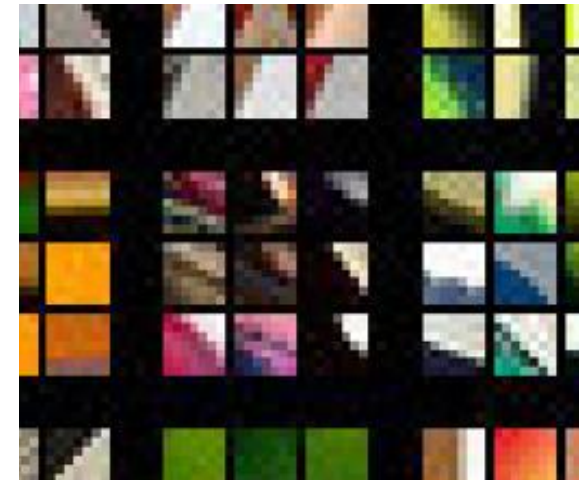


Image-level variation is high



Part-level variation is low

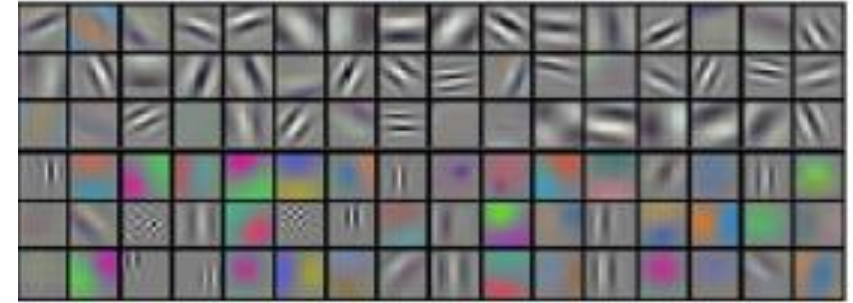
Image sources:

1. CIFAR 10 [CIFAR-10 image classification with Keras ConvNet - Giuseppe Bonaccorso](#)
2. Zeiler, Matthew D., and Rob Fergus. 2014. "Visualizing and Understanding Convolutional Networks." *ECCV 2014*

Visualizing CNNs

- CNNs are cool 😊 but some of the below questions need answers before we move forward :-
- How do I interpret the learned filters?
- What is it that stimulates/excites a neuron?
- How do I decide the architecture or improve existing ones?

Source: Krizhevsky et.al. NIPS'12



Visualizing the first conv. layer is possible but how about the later layers.

?

Composition of filters

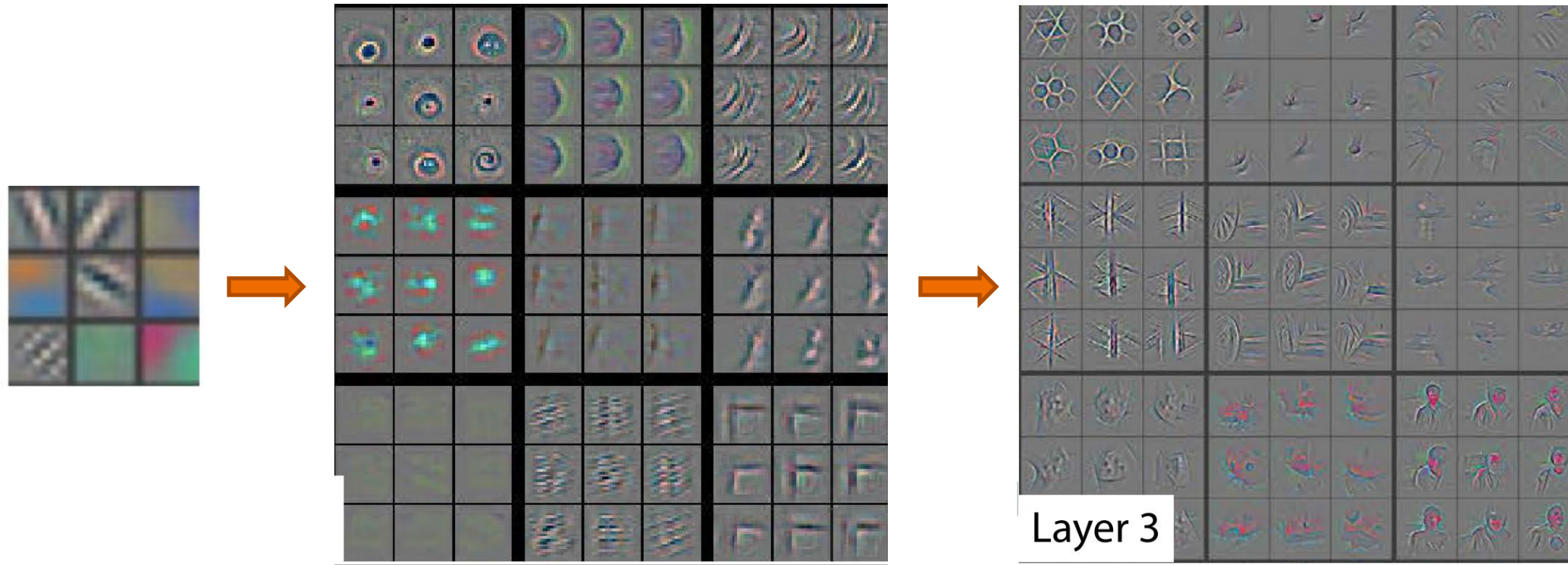
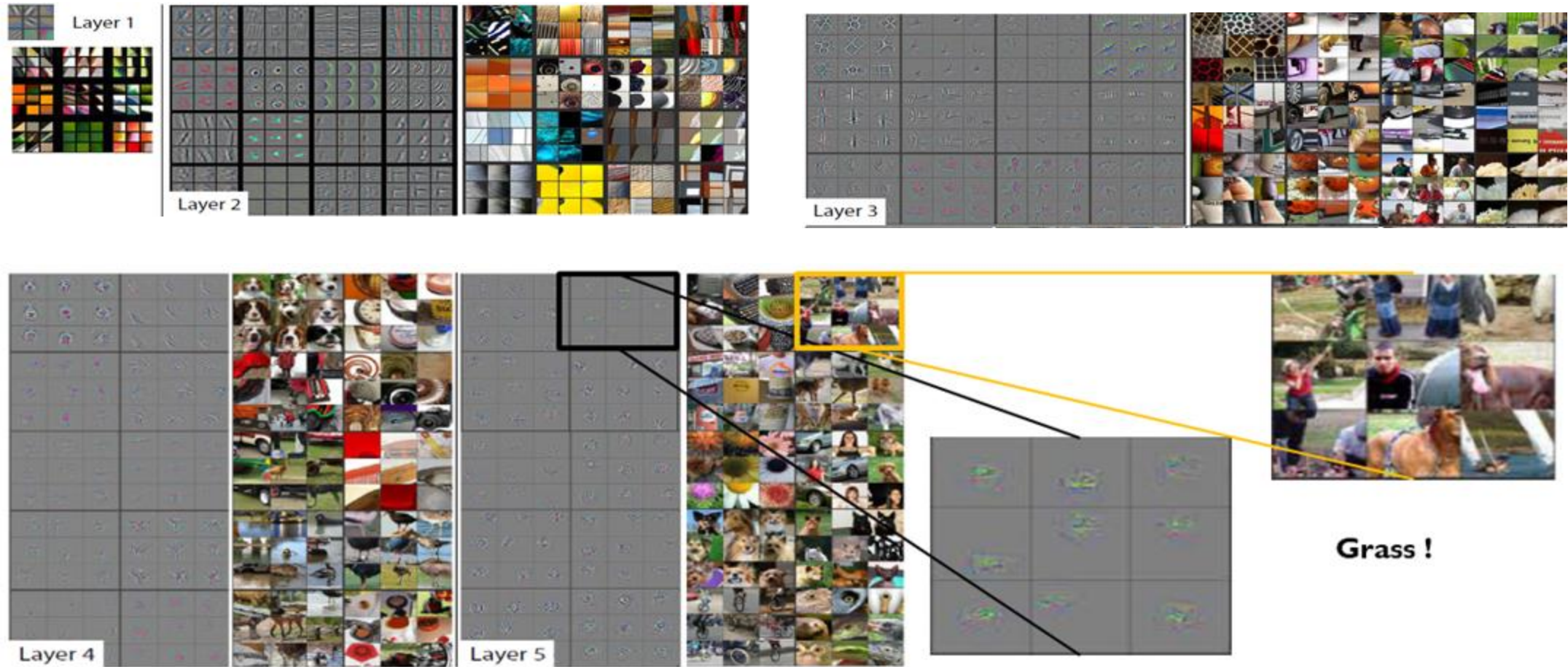


Image source:

Zeiler, Matthew D., and Rob Fergus. 2014. "Visualizing and Understanding Convolutional Networks." *ECCV 2014*.
These are top activations projected down to pixel space using deconvolution architecture

Visualizing CNNs

A. How do I interpret the learned filters?



Source: Zeiler e.t. al. ECCV'14

Early Layers Converge Faster

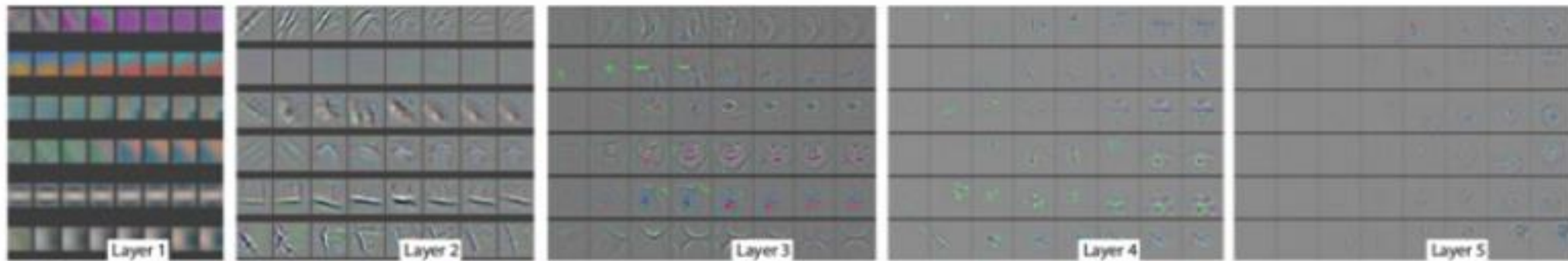


Figure: Evolution of randomly chosen subset of model features generated using deconvnet through training at epoch 1, 2, 5, 10, 20, 30, 40, 64.

Example: Classification of Documents

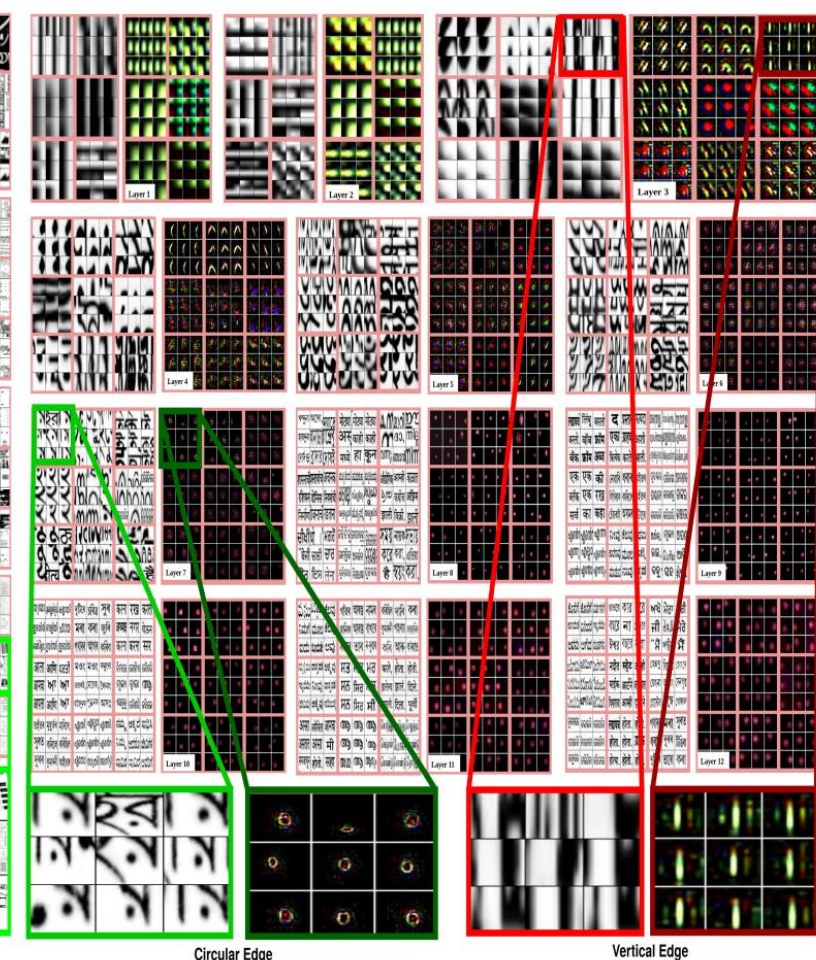
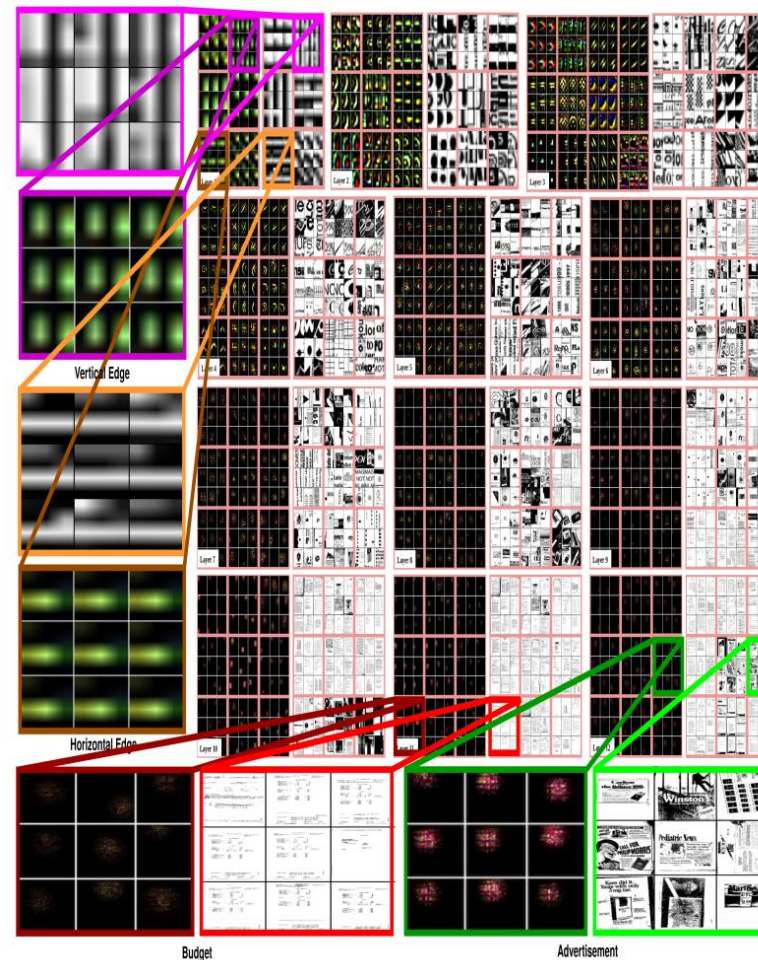


Task 1

Task 2

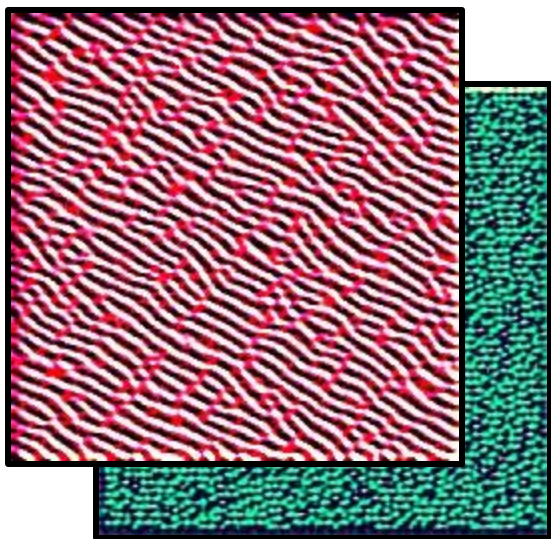
Task 3

Examples

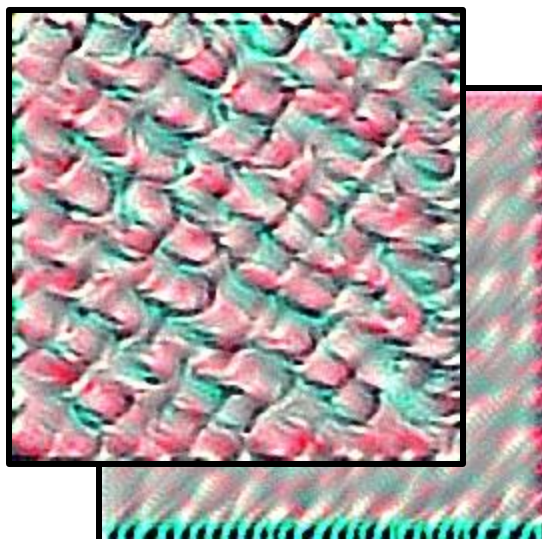


What does it look for faces?

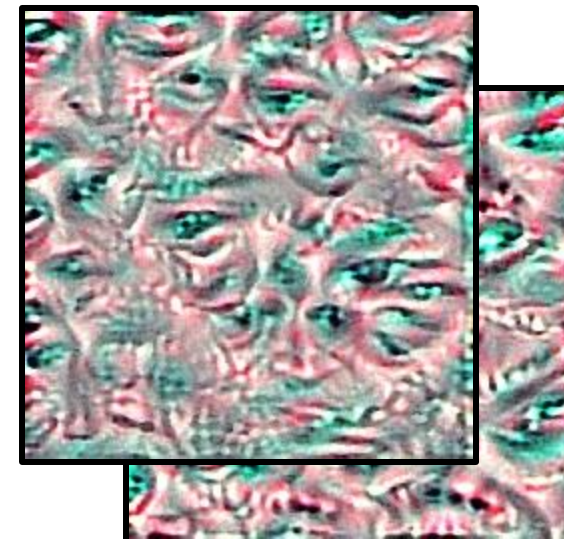
- What type of image causes a convolutional filter to give a high activation?



Low-level features

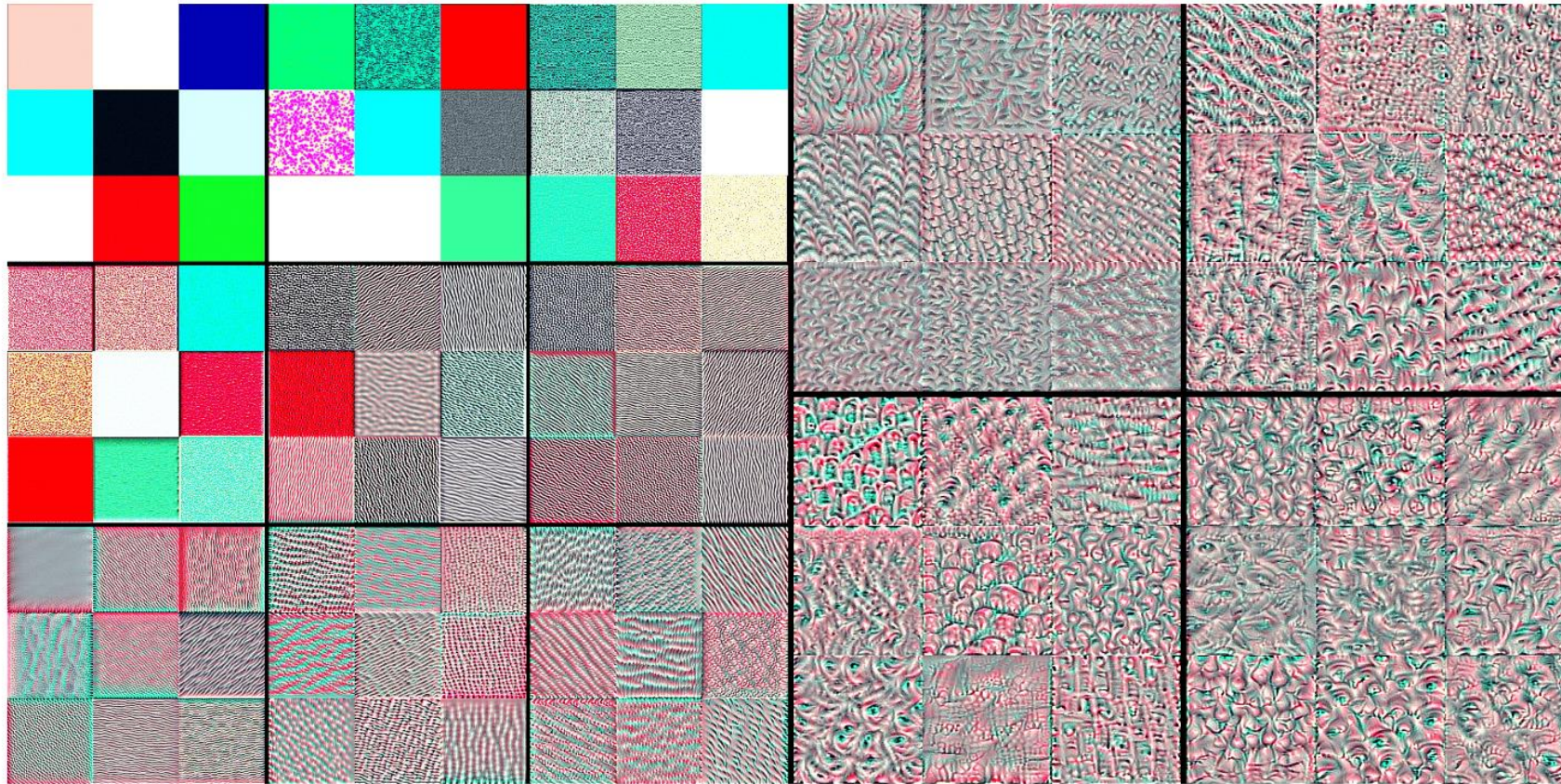


Mid-level features



High-level features

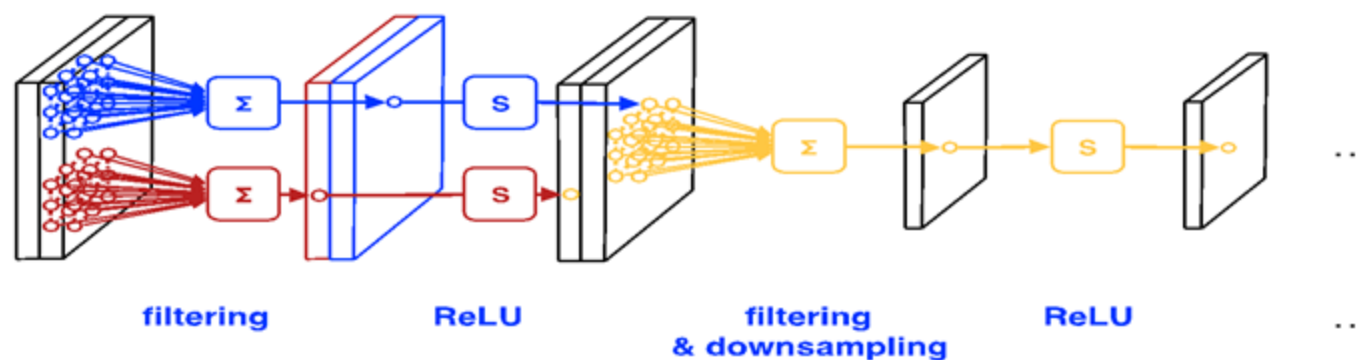
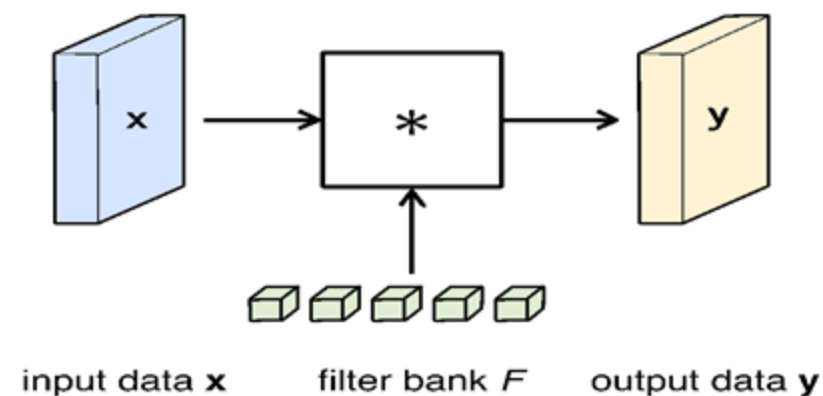
Results: face recognition



CNNs: Summary

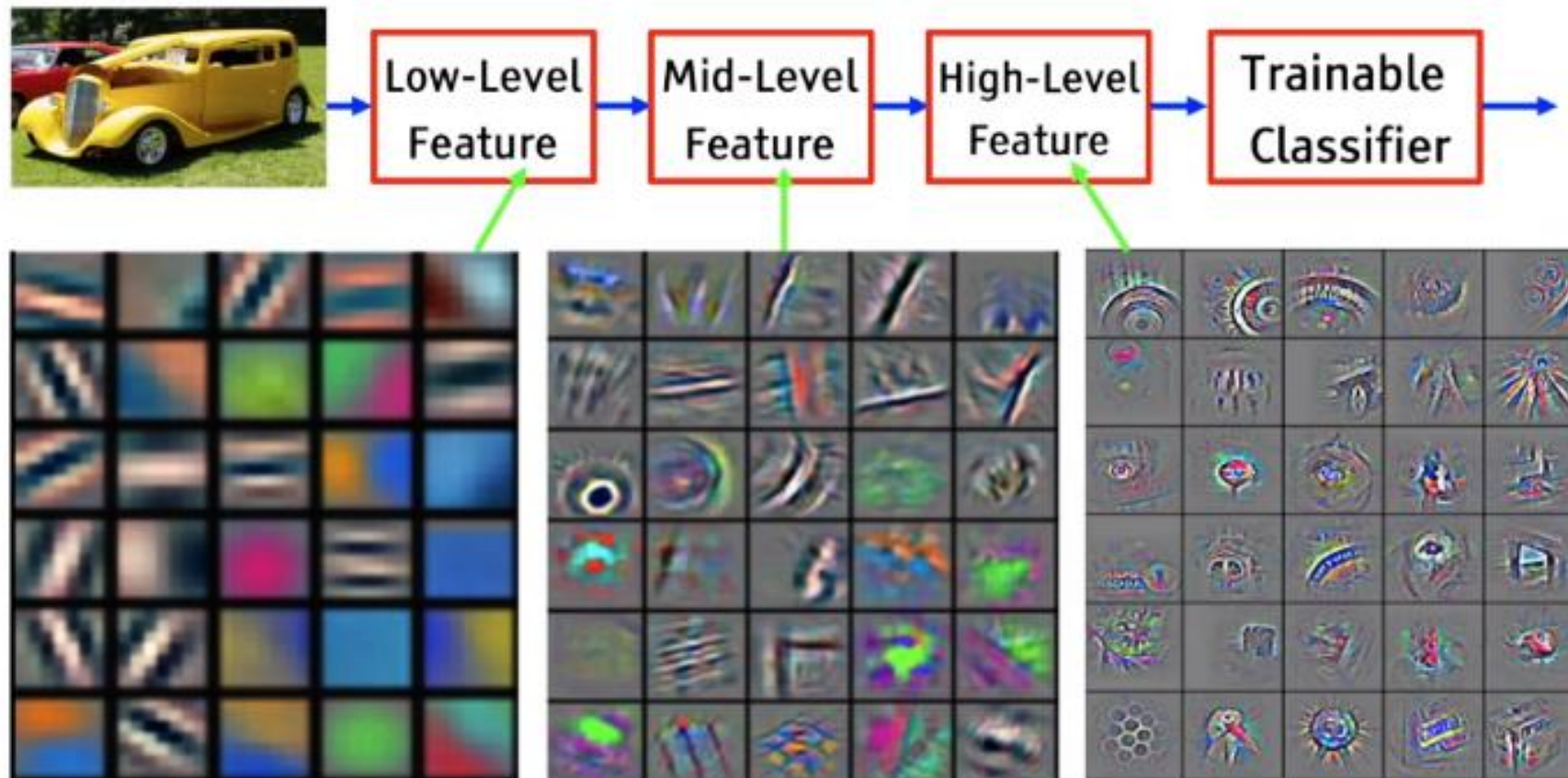


$$y = F * x + b$$

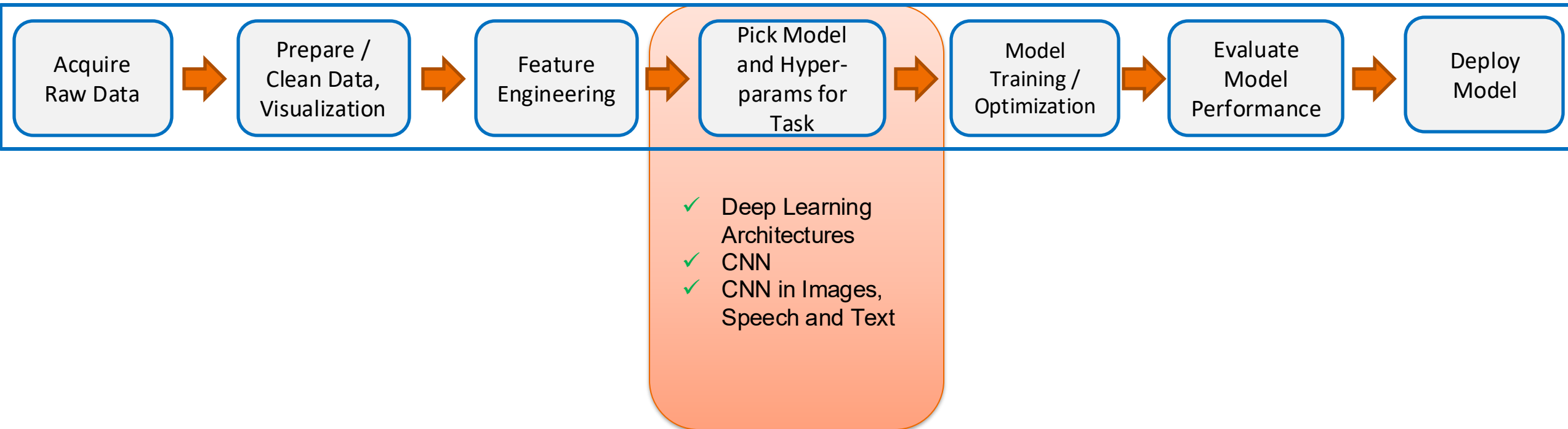


Deep Learnt Features

- It's **deep** if it has **more than one stage** of non-linear feature transformation.



Summary



Thanks!!

Questions?